

Phase 0 — Foundation (Week 1)

Goal

Set up the entire development infrastructure.

Tech Stack

Frontend

- Flutter
- Riverpod (state management)
- GoRouter (navigation)

Backend

- FastAPI
- Python 3.13

Database

- PostgreSQL
- Redis (caching)

Realtime

- WebSockets
- Firebase Cloud Messaging

Authentication

- JWT
- OAuth

Maps

- OpenStreetMap
- Mapbox SDK

AI

- TensorFlow
- Scikit-Learn

Cloud

- [AWS](#)

Phase 1 — Accident Detection Core

Goal

Detect crashes automatically.

Features

Sensor Monitoring Service

Collect:

- Accelerometer
- Gyroscope
- GPS

Flutter packages:

sensors_plus

geolocator

Crash Detection Engine

Detect:

- Sudden deceleration
- Sudden impact
- Device rollover

Logic:

Acceleration > Threshold

AND

Speed drops rapidly

AND

Gyroscope spike

SOS Countdown

Accident Detected

Are you safe?

15

14

13

...

No response:

Emergency protocol starts.

Backend APIs

POST /accident/report

POST /sos

GET /location

Phase 2 — Emergency Response Engine

Goal

Notify people immediately.

Features

Emergency Contacts

Store:

Mother

Father

Friend

Automatic Alert

Send:

Possible accident detected.

Live location:

Live Tracking

Realtime location updates.

WebSockets:

Victim App



Backend



Family App

Police Integration Layer

Future-ready API structure.

POST /authority-alert

Phase 3 — AI Severity Prediction

Goal

Predict seriousness.

Inputs

Impact force

Speed

Orientation change

Response delay

Location type

Model

Start:

Random Forest

Later:

XGBoost

Later:

Neural Network

Output

Critical

Moderate

Minor

and

Risk Score

0-100

Backend AI Service

/ai/severity

returns

```
{  
  "severity": "critical",  
  "score": 92  
}
```

Phase 4 — Trauma Center Intelligence

Goal

Find the best hospital.

Hospital Database

Store:

Hospital Name

Trauma Level

Ventilator Count

Emergency Beds

Specialists

Recommendation Engine

Formula:

Hospital Score

=

Distance

+

Travel Time

+

Trauma Capability

+

Availability

AI Chooses

Not nearest hospital.

Best hospital.

APIs

GET /best-hospital

Phase 5 — Volunteer Responder Network

Goal

Get help before ambulance arrives.

Volunteer App

Users register.

Store:

Training Level

Location

Availability

Smart Matching

Find:

Nearest 5 Volunteers

within:

1 km

3 km

5 km

Push Notification

Emergency nearby.

Can you help?

Volunteer Dashboard

Show:

Navigation

Victim location

Instructions

Phase 6 — AI First Aid Assistant

Goal

Guide helpers.

LLM Integration

Possible models:

- Llama 3
 - Gemma
-

Knowledge Base

Store:

CPR

Bleeding

Burns

Fractures

Shock

Voice Assistant

Speech-to-text

Text-to-speech

Languages:

Hindi

English

Marathi

Tamil

Bengali

Example

User:

Victim is unconscious

AI:

Check breathing.

Do not move the neck.

Call emergency services.

Phase 7 — Ambulance Intelligence

Goal

Optimize response.

Features

Ambulance Tracking

Realtime GPS.

Route Optimization

Use:

- Traffic
 - Distance
 - Road conditions
-

ETA Prediction

AI predicts:

Arrival in 7 minutes

Ambulance Dashboard

Shows:

Victim

Severity

Hospital

Best Route

Phase 8 — Government Dashboard

Goal

Centralized monitoring.

Features

Live Accident Map

Shows:

Active incidents

Analytics

Shows:

Response Time

Fatality Trends

Hospital Usage

Volunteer Performance

Heatmaps

Accident hotspots.

Phase 9 — Predictive Road Safety AI

Goal

Prevent accidents before they happen.

Inputs

Weather

Traffic

Past accidents

Road conditions

Outputs

High Risk Road

Moderate Risk Road

Safe Road

Driver Alerts

Warning

Accident-prone zone ahead.

Final Architecture

Flutter App

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API Gateway (FastAPI)

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Microservices

- └─ Accident Detection Service
- └─ Severity AI Service
- └─ Volunteer Service
- └─ Hospital Recommendation Service
- └─ Notification Service
- └─ Traffic Service
- └─ First Aid AI Service
- └─ Analytics Service

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PostgreSQL

Redis

Firebase

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AI Layer

TensorFlow

Scikit-Learn

LLMs



AWS Cloud

What to Build for the Hackathon (2–7 days)

Must Build

1. Accident Detection
2. SOS System
3. Live Location Sharing
4. Severity Prediction
5. Hospital Recommendation

Demo Only / Mock

1. Volunteer Network
2. AI First Aid Assistant
3. Ambulance Dashboard

This gives you a realistic MVP that can be built by a small team while still demonstrating the vision of the complete Golden Minute AI ecosystem.